CS4104 Computer Graphics & Virtual Reality

CIE3 CASE STUDY - RETAIL AND REMOTE COMMERCE

Student	Munnaluri Rishikesh
USN	1RVU23CSE291
Selected Case Study	Case Study 3 - Retail and Remote Commerce

1. Scenario

An interior-design startup wants to build a WebXR application that allows users to view furniture in their homes using an Augmented Reality (AR) or Virtual Reality (VR) environment. The system must make furniture respond naturally to room lighting, show realistic surface detail without excessive geometry, and map interaction space correctly to a phone or headset display.

2. Illumination & Color Models

Assignment question: When a user places a virtual red velvet chair into a brightly lit, sunlit room, what illumination models (ambient, diffuse and specular reflection) must the rendering engine compute to make the chair blend naturally with real-world lighting?

Ambient reflection represents general background light and prevents unlit portions of the chair from becoming completely black. **Diffuse reflection** models matte light and depends on the angle between surface normal N and light direction L. **Specular reflection** produces highlights using reflected direction R, viewer direction V and shininess exponent n.

Simplified Phong model: I = Ia*ka + Ip[kd*max(N.L,0) + ks*max(R.V,0)^n]. For velvet, diffuse reflection should dominate and specular intensity should stay moderate so the material does not look like glossy plastic. In WebXR, device light estimation can drive the virtual light direction and intensity when available.

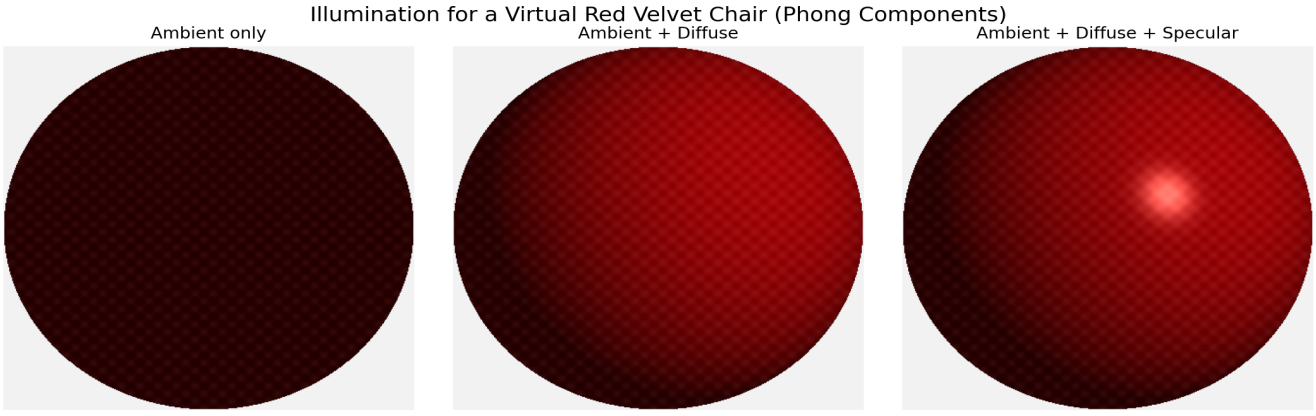


Figure 1. Ambient, diffuse and specular contributions on a synthetic red velvet material.

3. Texture & Bump Mapping

Assignment question: To make a leather sofa look highly realistic without increasing the polygon count of the mesh, discuss the application of texture mapping and bump mapping.

Texture mapping applies a 2D image to the 3D surface through (u,v) coordinates. A leather texture can store color variation, grain, seams and small imperfections. **Bump mapping** perturbs the surface normals used for lighting rather than changing the mesh geometry. This makes pores, grooves and stitching appear raised or recessed while polygon count remains unchanged. A normal map is a common practical form of bump mapping.

This is valuable for WebXR because a high-polygon sofa increases vertex processing, memory use and GPU load. A low/medium-poly model with texture and bump/normal maps provides a better realism-to-performance trade-off.

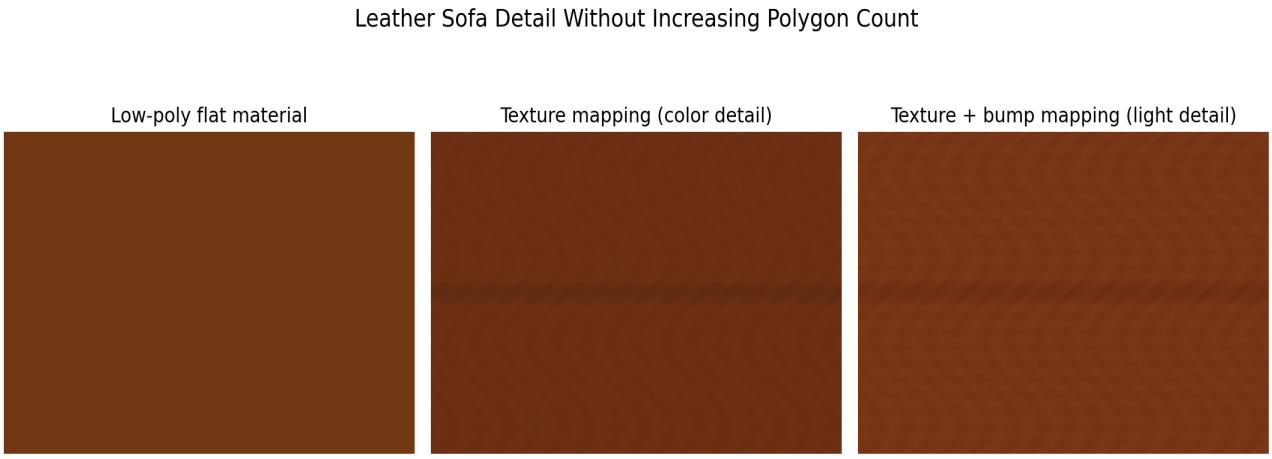


Figure 2. Surface detail added by texture and bump mapping without extra geometry.

4. Window-to-Viewport Transformation

Assignment question: How is window-to-viewport transformation used in the application UI to map the 3D interaction space accurately onto a smartphone or VR headset screen?

After projection, the interaction area can be represented in a logical window and mapped to device pixels. x is mapped by xv = xv_min + (xw - xw_min)(xv_max - xv_min)/(xw_max - xw_min), with the same form for y. This keeps selection handles, touch input and overlays aligned with the projected furniture.

In the demonstration, window = [-1,1] x [-1,1] and viewport = x:[70,1010], y:[120,1800]. The logical point **(0.25, -0.30)** maps to approximately **(657.5, 708.0) pixels**.

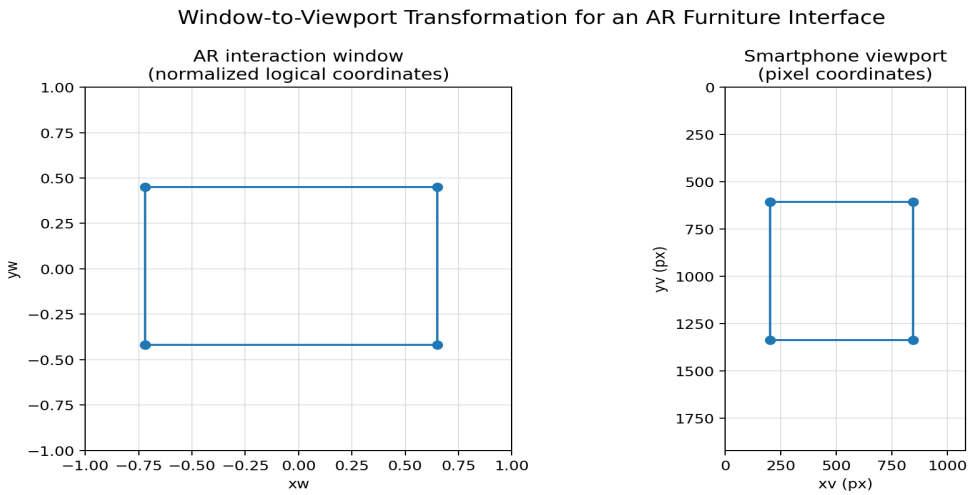


Figure 3. Mapping the same furniture interaction region from logical AR coordinates to a smartphone viewport.

5. Proposed WebXR Pipeline

Camera / VR tracking
Light estimation
Furniture 3D mesh
Texture + bump/normal maps
Ambient + diffuse + specular shading
3D viewing & projection
Window-to-viewport mapping
Phone / VR headset display

6. Optimization & Performance Analysis

For real-time WebXR, use low/medium-poly furniture meshes, compressed textures, mipmapping, a small number of dynamic lights, frustum/occlusion culling and level-of-detail models. Fine detail should be shifted to texture and normal maps rather than geometry. These choices reduce memory use and fragment/vertex processing while preserving visual quality and frame stability.

7. Practical Integration

The user opens the WebXR visualizer, scans the room, chooses furniture, places the 3D mesh, and moves/scales/rotates it. Texture and bump maps provide material detail, lighting parameters are adjusted to the environment, the viewing pipeline projects the object, and viewport mapping aligns UI controls with the display.

8. Result

The implemented demonstrations show that ambient, diffuse and specular components produce more convincing material appearance; texture and bump mapping add leather detail without extra polygons; and window-to-viewport transformation converts logical interaction coordinates into device-pixel coordinates correctly.

9. Conclusion

The case study demonstrates how core computer-graphics concepts directly solve a realistic WebXR retail problem. Illumination models make virtual furniture respond naturally to real-world light, texture and bump mapping provide efficient surface realism, and viewport transformation ensures accurate interaction on smartphones and VR displays.

Submission assets: Python implementation, executed Jupyter notebook, generated output images, Markdown report and this PDF.